

Heredity before individuality in a digital primordial soup

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Theories of the origin of life disagree about what came first but agree on the transition: a spreading pattern became an individual whose future depends on itself, not on its surroundings¹⁻⁴. No experiment has let it be watched and dissected. We did so in thousands of worlds of random machine code executing one another. Counting fixes the order: a copier that writes fewer symbols than it reads cannot finish a copy of itself in one pass, so it must run on into its surroundings or come back on itself. In every world of two real instruction sets the first replicator takes the first way: a two-byte word that writes itself when executed, whose offspring are exact copies in only half of its encounters. Evolution's first act is the second way: one jump that returns execution inside the organism. It wins within forty encounters per cell, makes the offspring one string in every context and inherits almost no variation, because it executes everything it copies. Genomes varying in bytes copied but never run appear and lose to smaller closers. In a second machine one instruction decides whether life begins open, and the lethality of sterile order how the open phase ends.

Main

What distinguishes a living thing from the chemistry around it has two answers: a list of properties, metabolism, compartment and heredity, with the question of which came first⁵⁻⁹, or a relation: a living system is one whose operations produce the organisation that produces them, so that its future depends on its own past rather than on its environment, in Maturana and Varela's operational closure¹, Rosen's closure to efficient causation², Fontana and Buss's organisational closure^{3,10} and, made quantitative, the information-theoretic individuality of Krakauer and colleagues^{4,11}. This is also the moment at which population genetics places the start of Darwinian evolution proper, when vertical descent comes to dominate communal exchange^{12,13}. No experimental system has allowed this transition to be observed as it happens, repeated under controlled variation, and taken apart.

Digital organisms usually descend from a designed ancestor^{14,15}; soups of random programs come closer. When random byte strings for a simple machine execute one another, with no fitness function and no replacement, self-replicating programs arise and take over¹⁶, as they did in the Coreworld^{17,18}, among random Lisp expressions¹⁹ and in artificial chemistries^{20,21}. For the Z80 processor the first replicators are known to be load-push programs that give way to a block-copy family^{16,22}, a takeover attributed to mutational robustness²². We treated such a soup as a test of one explanation. A copier that writes fewer symbols than it reads cannot complete a copy of itself in one pass over its own cells, so its first pass must end either by running on into whatever lies beyond it or by returning to a cell it has already executed. If life begins with the smallest thing that writes itself, it must begin open, and individuality must be a later invention that closes the path of execution. We ran more than 3,600 worlds of random Z80 code, varying the instruction set, program length, execution budget, mutation rate and memory geometry, with confirmatory stages pre-registered with kill criteria (Methods). Heredity is measured by intervention, as in serial-transfer tests of self-replication²³: a program is lifted out and cultured against fresh random partners, so that a pattern that merely spreads is not mistaken for one that inherits. We then repeated the central experiment on the Z80 restricted to its 8080 ancestor's instructions, and on a second, minimal machine whose instruction set we could edit.

Order before life

The soup holds 20,000 programs of L bytes on a 160×125 lattice (Fig. 1a). At each step 8,192 random cells are each paired with a lattice neighbour, the pair is executed as one program for 128 instructions and written back, and a background mutation replaces a fixed number of random bytes; nothing else happens (Methods). A tape is *heritable* if, lifted out and executed against random partners, it makes copies that copy in turn; emergence is dated by the first heritable tape among the most common, and the heritable fraction of random cells is the population measure.

The first thing to form is order that does not inherit. When the most common pattern first occupies a tenth of the soup, 24–34% of all bytes are zero at every tape length from 9 to 100 (Fig. 2). The zeros are self-written: push, call and reset

instructions executed with empty registers write zeros wherever the stack pointer points, and zero is itself an instruction that does nothing, so the pattern spreads by being written and changes nothing when executed. We call this flood *tar*, after the asphalt that prebiotic chemistry makes from the reactions that also make sugars²⁴. The tar is what detectors see: the high-order entropy used to detect the transition in random-program soups¹⁶, an excellent detector at $L = 16$, is near chance or inverted in unablated worlds at six of the eight lengths from 25 to 100 bytes (area under the curve 0.07–0.54), a complexity-times-abundance measure of selection^{25,26} rises nineteen-fold before the first heritable replicator exists in 71 of 80 worlds, and the assembly index of the first replicator equals that of the modal sterile tape in 79 of 80 (Extended Data Fig. 1). Order comes first, it is as simple as what follows, and only intervention tells the two apart.

The first replicator is open

The first heritable replicator is the same program everywhere: the two-byte word 01 c5, or one of its register variants, repeated to fill the tape (Fig. 1a). It is the first replicator in 170 of 170 worlds of the confirmatory and control stages with benign tar, across five tape lengths, two instruction sets and two pairing rules, at medians of 325–750 steps on the lattice whatever the length and 125 steps with well-mixed pairing (Fig. 1b). Executed, the word loads its own next two bytes into a register pair and writes that pair where the stack pointer points, in the partner: the program is its own data, and it copies itself as a side effect of being run. It has no loop, no counter and no test of its surroundings: the ring returns the pointer, and every byte of the tape is the same word. It is also nearly the only thing of its size that replicates: of all 65,536 two-byte words tiled to 16 bytes, exactly five copy themselves into an all-zero partner, all of them load-and-push pairs (Extended Data Fig. 2; Supplementary Information, Proposition 3). Removing the instructions it uses removes it: without immediate loads 2 of 10 worlds produce a replicator by 300,000 steps, without the stack writers life arrives 160- to 992-fold later as a block-copy replicator, and every other ablation leaves emergence within seed-paired noise (Extended Data Fig. 3).

The first replicator is *open*. Its pointer fetches partner bytes in every one of 256 encounters, for every first replicator under benign tar (100 of 100 tapes on a traced copy of the executor; Methods). Against 256 random partners it copies into a median 0.66–0.73 of them and loses a quarter or more of its own bytes in a median 0.06–0.43 of encounters (Fig. 3a,b), because after writing its copy it executes the partner's bytes, which can poison the register pair before the next push. With the organism fixed and execution deterministic, the information that flows from the partner into the offspring equals the entropy of the offspring over random partners (Methods). The offspring is an exact copy of the parent in a median 56% of encounters at $L = 16$, 29% at $L = 20$, 5% at $L = 50$ and 0.8% at $L = 64$; every one of its L positions varies across partners; and its entropy over 256 partners runs from 3.9 bits at $L = 16$ to the 8-bit ceiling of the sample at $L = 64$ (Fig. 3c): what a child looks like depends in part on the neighbour it was made against. In the population the heritable fraction of random cells stays between 0.07 and 0.19 for the first 20,000 steps after emergence at $L = 16$ (Fig. 3d): the first replicator's reign is a world of damaged copies.

Closure wins the competition and inherits nothing

The first replicator does not stay. At $L = 16$ the dominant tape carries a control-flow instruction in 20 of 20 worlds by 300,000 steps, 17 of them ending on the identical eight-byte sequence whose conditional return sends the pointer back into the organism's own body; at $L = 50$ all 20 worlds carry a relative jump inserted into the pusher tiling, 14 of them byte-identical (Fig. 3e; Extended Data Table 1). Every such successor copies into 1.00 of 256 random partners with 0.00 self-damage, whichever instruction implements the cycle, a return, a jump or a hardware block-copy loop (Fig. 3a,b,f), and its offspring carry nothing detectable from the partner: in 63 of 67 such worlds the offspring is one and the same string in all 256 encounters (Fig. 3c). The heritable fraction of random cells rises to 0.75–0.94 when the closed successor takes over (Fig. 3d).

We call this property *closure*. Three things travel together here and must be kept apart: a loop in the code, which is syntax; confinement of the pointer to the organism's own cells, which is the mechanism; and independence of the offspring from the partner, which is the outcome. All three were measured. On the traced executor every loop-bearing final dominant fetches no partner byte in any encounter (89 of 89 across the confirmatory and control stages); confinement and zero inflow agree in 106 of 110 finals, the four exceptions being confined finals that keep one to three partner bytes, and no final whose pointer enters the partner is free of inflow (Methods; Extended Data Fig. 4). In this operational sense closure is operational closure¹ made countable and one component of individuality⁴ made measurable by intervention. It is not a membrane, since the organism's bytes are as exposed as before, and not a property of place²⁷: well-mixed pairing left the open phase as heritable and closure later rather than sooner (7 of 10 worlds against 20 of 20; Methods). It is a cycle in the organism's control flow that never leaves the organism (Fig. 3f). The return and the block copy need nothing but the instruction set;

the relative jump depends on a convention of the harness, a 16-bit program counter in a memory addressed modulo $2L$, which lands a backward jump across address 0 inside the organism only at $L = 20$ and 50 , and 25 of the 67 closers use it (Methods). At the aligned lengths, where no jump can use the wrap, closure is slower and monotone in L : 20 of 20 worlds by 300,000 steps at $L = 16$, 12 of 20 by one million at $L = 32$ and 8 of 20 at $L = 64$, all by block copy; given ten million steps, every world closed at $L = 16$ and 20 (20 of 20; Extended Data Fig. 5).

The return is what selection acts on. Seeded at 1% into a world filled with its own ancestor, the pusher with its jump holds half the cells within 110–130 steps in 5 of 5 soups, about forty encounters per cell, and the ancestor seeded into a world of the closed form leaves no trace; in pusher-filled worlds with no seeding the jump arises by mutation and sweeps within 5,000–13,250 steps in 3 of 3 (Fig. 4c; Methods). The advantage measured in isolation, 1.00 against 0.69 of partners copied, is a competitive advantage among kin.

It is paid for in variation. A pre-registered single-byte mutational scan of all 160 first and final tapes shows that closure did not make the organism a better carrier of inherited differences (Fig. 4b; Extended Data Fig. 6). The first replicator is at least as robust to mutation as its successor at every length (a median 0.74 against 0.59 of single mutants remain heritable at $L = 16$, 0.99 against 0.95 at $L = 50$), so robustness²⁸, the explanation offered for the same succession elsewhere²², does not account for it. And the first replicator transmits its mutations: a changed operand byte is pushed into the copy and pushed again by the copy, at a median 23 of 50 and 32 of 64 positions. The closers transmit almost none, a median 1 of 50 positions for the pusher with its jump and 0 for the return and block-copy designs, because what they copy is the code they execute, the rest of the tape being regenerated from it, so a change is either lethal or erased. The first closed organisms are one string in every context and one string across their own mutants.

The variation that remains lives elsewhere. The pusher's transmissible positions are all bytes it executes, operands read as instruction stream and pushed as data; the closers' few are bytes they copy but never execute (Fig. 4a), 82 of 518 sites among the final dominants and 20 of 24 among those born under lethal tar: a genotype in von Neumann's and Langton's sense^{29,30}, copied as data and not read as code. The channel does not reopen: in ten-million-step runs the most common tape at the end is a minimal closer with no transmissible site in 17 of 20 worlds, and the pre-registered recovery criterion failed at both lengths (1 of 10 and 0 of 10 worlds; Methods). Genomes do arise: in 3 of 20 worlds an irregular block copier carrying 10–12 transmissible sites, 7–10 of them never executed, was the most common tape for up to two million steps, never above a third of a percent of the soup, and each was displaced within five million steps by a design carrying fewer sites or none (Fig. 4d). The first individuals stay canalised³¹ for as long as we watched.

Two properties of the substrate decide the beginning

If the order of events is forced by the literal channel, a machine without one should be born closed, and giving it one should start life open. BFF, the Brainfuck family of Agüera y Arcas et al.¹⁶, has no literal channel: its instructions increment in place or copy from memory, and no execution of fewer than 129 instructions can copy a 64-byte tape with two or more distinct bytes (Theorem 1). Soups of 2^{17} programs run for 16,384 epochs test the prediction (Fig. 5). In BFF as published and under a wrapping pointer, 28 of 48 worlds produced a heritable replicator; every first replicator contains a loop (28 of 28) and none is open; with harmless brackets, 7 of 12 did, all loop-bearing, and four lost the replicator again by 16,384 epochs. Adding one instruction that writes its own two operand bytes below the write head changes the beginning completely: in 12 of 12 worlds the first replicator is the one-byte tiling of that instruction, loop-free, with its pointer entering the partner in every encounter. It is the 01 c5 of BFF, and its offspring carry 8 bits of the partner, the ceiling of the measurement (Extended Data Fig. 7).

Then it dies: within about 200 epochs each soup fills with tar that halts the pointer, since an unmatched bracket ends BFF execution; copies into such partners are cut short and sterile, the open replicator persists only as a 0.2% minority, and no closed class evolves, none existing in this language up to period 10 (Methods). The open phase is a *window* whose length the tar sets. Making an unmatched bracket a no-op leaves the open replicator in place in 12 of 12 worlds for all 16,384 epochs (Fig. 5c); making it halt with probability p reveals a threshold rather than a dial: the replicator holds 98% of the soup at $p \leq 0.1$ (18 of 18 soups), and at $p = 0.3$ it collapses within about 900 epochs and recovers only to a tenth of the soup (6 of 6; Fig. 5d). The write ratio sets the other boundary. When the literal instruction writes two or three copies of its word instead of one, so that a copy completes within one pass, the first replicator is still the loop-free tiling whose pointer enters the partner in every encounter, but its offspring carry 0 bits of the partner, information-closed from birth without a cycle, and, unlike at the published ratio, it never collapses: every random tape is heritable at 16,384 epochs in 12 of 12 soups (Fig. 5d, right). The lethal column holds in the first machine too. Making a zero byte halt execution, so that the Z80's own tar becomes lethal, removes the open beginning altogether: in 10 of 10 pre-registered worlds no literal pusher ever establishes, and life begins closed and late, as a block-copy replicator with a hardware loop, at a median of

46,250 steps against 525 under benign tar (Extended Data Fig. 8). And the beginning is not the Z80's alone. Restricting the Z80 to the Intel 8080's instructions, by making every Z80-only instruction a no-op, keeps the literal word and removes every closer but the return (Methods). Three predictions were registered and met: the load–push word comes first in every world; at $L = 16$ the byte-identical return design, pure 8080 code and the Z80's own modal closer, dominates 20 of 20 worlds by 300,000 steps; at $L = 32$, where the return's popped addresses fall in the partner and there is no block copy, none of 20 worlds closes by one million steps and the open pusher reigns throughout (Extended Data Fig. 5), which is also why long 8080 tapes never produced a looping variant in ref. 16. The beginning is the machine family's; what follows depends on whether the instruction set can express a closer at that ring size. Together these fill a map (Fig. 5e): a literal channel decides whether life starts open; the lethality of the tar decides whether an open beginning lasts, or happens at all; and the supply of closed designs decides whether what follows is closure, a late closed beginning, or extinction.

What is proved

Two counting statements make the order of events a theorem rather than an observation (Fig. 6; Supplementary Information). A straight-line program's pointer runs on into its partner, so a straight-line replicator is open by construction; and a closed replicator, whose pointer never leaves its own cells, must execute some address twice, because its 128 executions per encounter outlast a pass over at most 100 cells and nothing stops execution inside it, or, in BFF, because a copy costs more executions than the pair has cells: closure requires a cycle (Theorem 2; Fig. 6a). A straight-line replicator exists wherever the instruction set can write a literal at a self-advancing pointer, and it is the size of the literal; in this machine no closed design is as small (Proposition 3; Fig. 6b), so the open form is present in a random soup with exponentially higher probability than any closed one³². Without a literal channel the count goes the other way: in BFF every replicator carries a loop (Theorem 1). The boundary of Theorem 2 is itself a prediction, confirmed twice in the second machine: a copier that writes at least as much as it reads closes by saturation without a cycle, as the wrapping literal pusher does (0.04 bits) and as the literal that writes two or three copies of its word does with no wrap at all (0 bits), each entering its partner in every encounter yet importing nothing from it. In information terms closure is zero inflow from context to offspring, $I(o; e | x) = 0$, which perfect fidelity requires (Lemma 7), and Fig. 3c shows the measured inflow reaching zero as fidelity reaches one. The dynamics lie outside the theorems: closure arrives with probability about $1 - \exp(-q \int n dt)$, q being the rate of closing mutations and n the open population, so benign tar lets the window run until a closer is found and lethal tar shuts it first (Supplementary Information, Model 5).

Discussion

The order of events follows from counting once a machine's default flow and copy cost are given, and the smallest thing that writes itself has no room for a return. So the first replicator is open; closure is the return, reached by four different instructions and selected within forty encounters per cell; and the first individuals are canalised, while genomes that carry variation in bytes they copy but never run arise and, within ten million steps, lose to smaller closers. The explanation has few adjustable parts: the dichotomy is arithmetic, the literal channel is read off a substrate, and the lethality of by-products is measured. It answers questions it was not built for: why the first replicators are sloppy; why a cycle, not a membrane, is the first thing selection builds; why the capacity for inherited variation can fall at the first transition in individuality; and why the first innovations after replication should concern where copying starts, stops and returns. When the literal road is blocked, by removing the stack writers, by a language without a literal channel or by tar that halts whatever executes it, life still comes, late and born closed: the exponentially rarer closed beginning, visible as a ninety-fold delay.

These are predictions for chemistry, not results about it. Template copying is a literal channel, a strand specifying its complement by physical fit with no interpreter, so the first replicators of such a chemistry should be the shortest self-templating repeats, dependent on their surroundings and productive mainly of partial copies; transfer into fresh material and the variation of the product with its context would measure both. By-products are a variable: whether a chemistry's tar is inert or inhibitory to copying should decide whether an open phase lasts long enough for control to evolve, which matched inert and inhibitory additives in nonenzymatic systems can test²⁴. Processivity is the chemical write ratio, which in the second machine closes a copier in information without any control: vertical lineages should appear when one copying event spans the genome³³, and in laboratory evolution of RNA replicators the earliest fixed changes should concern control, circularisation, terminators and primers^{34,35}, with inherited sequence that is copied but not used rare and transient at first. Detection of life at its origin should be by intervention, since low-complexity order precedes the first replicators and trips compression and assembly measures alike^{25,26}. What limits the claims is plain: two real instruction sets and one toy language; units of fixed size, start address and pairing imposed by the harness; a culture test that is our own ground truth, though the order of events held under every setting we tried; and closure at the largest lengths in progress rather than

complete within our horizons. Within those limits the result stands: heredity came first; independence from context came later, as a return in control flow that cost the organism its capacity to vary; and the step that made a replicator an organism was the step that made its offspring stop depending on its environment.

Methods

Soup. The soup holds $N = 20,000$ tapes of L bytes (L from 3 to 100), initialised with uniformly random bytes, on a 160×125 square lattice. At every step 8,192 cells are drawn uniformly at random and each is paired with one of its four lattice neighbours (a step off the edge is reflected; the lattice does not wrap); a pair is active if neither member is already in another pair that step, which leaves $\approx 3,980$ active pairs. An active pair is concatenated, organism first, into a pair memory of P bytes ($P = 2L$ except in the padding experiment of Stage F, where $P \geq 2L$ and the padding bytes are never written back), executed by a Z80 emulator for a fixed budget of 128 instructions (32–2,048 in the budget arms) from byte 0 with all registers zero and the stack pointer at the last byte of the partner, and both tapes are written back. After each step $8,192/2^k$ bytes chosen uniformly at random over the soup are replaced by random bytes ($k = 4$ unless stated; $k = 1$ –6 in the mutation arms). There is no fitness function, no replacement and no death. Instruction-set ablations replace whole opcode families by no-ops through a prefix-aware fetch hook that sees the operation after DD/FD and ED/CB prefix resolution; the ablations and their opcode counts are listed in Extended Data Table 2. The emulator was validated against an independent Z80 reference with zero divergences in randomised 5,000-pair tests with and without ablation; no claim depends on cycle-exact timing.

Recording. Every run writes a sample at fixed steps (every step for the first 30, then a geometric schedule, then every 500 or 1,000) with the ten most common exact tapes and their shares, quasispecies³⁶ occupancy (fraction of cells within Hamming distance $\lceil L/4 \rceil$ of the dominant tape), byte histogram, zero-byte fraction and byte entropy, brotli-compressed bits per byte and the high-order entropy of ref. 16, species entropy and unique-tape fraction, per-interaction statistics (bytes written into the partner, copy events, destroyed programs), 16 random tapes for the functional census, and full soup snapshots at fixed steps and at emergence.

Culture test and partner tests. A tape runs as the first program against 64 uniformly random partners for the condition's budget; copy success is the fraction of partners that become $\geq 75\%$ copies at the best cyclic shift; gen2 is the copy success of the produced copies against fresh partners, weighted by the copy fraction; heritable means $\text{gen2} \geq 0.3$, faithful additionally $\geq 50\%$ of partners copied. Partner-independence tests use 256 random partners and record self-damage ($\geq 25\%$ of the organism's bytes changed in one encounter). In BFF the culture test also records whether the instruction pointer entered the partner in each encounter; closed means $\leq 5\%$ of encounters entered and $\geq 95\%$ of partners copied, open means $\geq 50\%$ entered. Information inflow from the environment, $I(o; e | x) = H(o | x)$ for a deterministic executor, is the plug-in entropy in bits of the offspring (the partner's bytes after the encounter) over 256 random partners, ceiling 8 bits.

Events and statistics. t_{rep} is the first sampled step at which one of the three most common tapes has a share $\geq 0.5\%$ and is heritable by the culture test (a clonality criterion by design: it dates the first *clone*, not the first heritable material); t_{faith} additionally requires faithfulness; $t_{q_{10}}$ (quasispecies occupancy $\geq 10\%$) is the pre-registered occupancy event and fires on zero floods as well as on replicators. Emergence fractions are quoted as worlds reaching the event within the horizon with Wilson 95% intervals; times as Kaplan–Meier medians with censoring at the horizon. Ablation effects are ratios of Kaplan–Meier medians with seed-paired sign tests (the same seeds run under every ablation); confirmatory contrasts use one-sided Fisher exact tests and the Cochran–Mantel–Haenszel test pooled over budgets; the paired gain of control flow uses the exact McNemar test; between-variant transition rates in BFF use two-sided Fisher tests and Mann–Whitney tests on event times with non-transitioning runs censored at the horizon. Across the 233 unablated worlds of the exploratory stages, 92 gained a control-flow instruction between the first replicator and the final dominant and 3 lost one (exact McNemar $P = 7 \times 10^{-24}$). In BFF, where replicator populations are quasispecies with small exact-class shares, the first replicator is the most common tape class (a tape and its reverse count as one class) at the first sample at which it is heritable, and the population event is a heritable fraction ≥ 0.5 of 32 random tapes. All thresholds (0.75 copy, $\text{gen2} \geq 0.3$, 0.5% share, 10% occupancy, 95%/5% closure) were fixed before the confirmatory stages. Same-seed Z80 runs diverge because GPU scheduling makes the pairing order non-deterministic, so every Z80 claim is a population claim over seeds (ten same-seed repeats spread as widely as ten seeds); the BFF soup is bitwise reproducible.

Definitions and theorems. A *self-writer* is a tape that, executed against the all-zero partner, leaves the partner holding a copy of itself. *Closure*: the organism's bytes are forward-invariant under execution for every context in the class of uniformly random partners; equivalently its pointer never executes a partner address. Theorem 1: any execution that copies a 64-byte BFF tape runs at least 129 instructions, so every BFF replicator executes some address twice, that is,

carries a loop. Theorem 2: in any machine whose pointer advances by default, a closed full replicator executes some address twice, provided a copy costs more instruction executions than the organism has addresses (BFF) or execution cannot stop inside the organism before the budget is spent (the Z80 soups, budget $128 > L$). Theorem 6: a self-consistent literal-write instruction gives a straight-line, hence open, replicator the size of the literal; without one, under the copy-cost condition, every replicator contains a cycle. Lemma 7: $I(o; e | x) > 0$ implies imperfect fidelity across contexts; perfect fidelity implies $I = 0$. Proposition 3 (computer-assisted): the two-byte words fixed under their own execution against the zero partner are exactly the five load–push words, all straight-line. Full statements and proofs are in the Supplementary Information.

Traced execution. A derived copy of the culture-test executor records every address fetched as instruction stream (opcodes, prefixes, operands) in a 256-bit map per encounter, with byte-identical execution (checked). Pointer confinement is the fraction of 256 encounters with a seeded random partner in which no partner address is fetched; the executed set is the union of fetched organism positions; a transmissible position of the mutational scan is called unexecuted if it is never fetched. Every first replicator and final dominant of Stages G, K and I (110 of each) was traced.

Mutational scan. For each Stage G first replicator and final dominant, every single-byte mutant (all 255 alternative values per position at $L = 16$ and 20 ; 32 seeded values per position at $L = 50$ and 64) was run as organism A against 32 shared random partners for 128 instructions and its offspring against 32 fresh partners, with the culture test’s thresholds. A mutant is heritable if $\text{gen2} \geq 0.3$; its mutation is transmitted if the mutant byte is present at the aligned position in at least half of the offspring that are $\geq 75\%$ copies; a position is a transmissible site if at least half of its tested values are both heritable and transmitted; capacity is $\sum_i \log_2(1 + 255 f_i)$ over positions, f_i being the fraction of transmissible values at position i . Three predictions were written before the run (the successor more robust to mutation than the first replicator; the closed form with more transmissible sites; the matched pusher-with-jump of $L = 50$ with more sites than its pusher); all three failed and are reported as such.

Pair invasion. Soups of 20,000 cells at $L = 50$ under the Stage G dynamics, every cell set to one tape of the matched pair (the pusher tiling, or the same tiling with $20 \text{ } f0$ inserted three times, the final dominant of 14 Stage G worlds) and 1% of cells to the other at step 0; five seeds per direction, three seeds per resident with no invader; 20,000 steps. The pre-registered outcome, the invader’s nearest-class share passing 50%, could not be met at this length because the quasispecies cloud splits between the two classes (a closed tape that has lost two jumps is nearer the pusher); the share of cells carrying the jump word, reported instead, was defined after the data were seen.

Address mapping and aligned lengths. Pair memory is a ring of $2L$ bytes: every memory access uses the 16-bit address modulo $2L$, and the stack pointer starts at the largest 16-bit value that aliases the last byte of the partner. The program counter and the stack pointer themselves are 16-bit, so a relative jump or a stack move that crosses address 0 reaches $(65,536 + \text{target}) \bmod 2L$, which equals the true-ring target only when $2L$ divides 65,536. Of the 67 closed final dominants of Stage G, 25 (all 20 at $L = 50$ and 5 of 19 at $L = 20$) use a relative jump whose target depends on this aliasing; the pair-invasion and mutational-scan results at $L = 50$ are results about the machine so defined. Stage K ($L = 32$, 20 worlds, one million steps, seeds 5001–5020; pre-registered: first replicator a load–push word in ≥ 18 of 20, a closed successor in ≥ 15 of 20, kill below 10) tested closure at an aligned intermediate length: 20 of 20 and 12 of 20, all closers block copies.

The 8080 subset. The `i8080` suppression makes every Z80-only instruction a no-op: the 256 CB-page opcodes, the 78 defined ED-page opcodes (block copies and their repeats, 16-bit ADC and SBC, IN and OUT (C), NEG, RETN and RETI, IM, LD A,I and A,R, RLD and RRD) and EX AF,AF’, DJNZ, the five JR forms and EXX. Two deviations from a real 8080 are kept: IX and IY prefixes still select the indexed form of an 8080 instruction, and suppressed bytes are no-ops rather than the 8080’s undocumented aliases. Stage M: Stage G conditions at $L = 16$ (300,000 steps) and $L = 32$ (one million), 20 worlds each, seeds 7001–7020, with the three predictions registered before the run.

Ten-million-step runs. Stage L: the Stage G conditions at $L = 16$ and 20 , ten worlds each (seeds 6001–6010), run for ten million steps with snapshots every 5,000 steps. At every snapshot of these and of the Stage G, K and I worlds, the most common exact tape was scanned (32 values per position, 32 partners) and traced (64 partners), giving its transmissible sites, capacity, executed set and pointer confinement against step. Predictions registered before the run: L1, in at least 5 of 10 worlds at each length the dominant at ten million steps has more transmissible sites than the closed dominant at 300,000 ($L = 16$) or one million ($L = 20$) steps; L2, every such recovery is in a block-copy lineage and its sites are copied-but-unexecuted positions; L3, no return to an open dominant. Outcomes: L1 not met (1 of 10 at $L = 16$, 0 of 10 at $L = 20$); L2 not met at $L = 16$ (the one recovery, from 0 to 4 sites, is a block-copy lineage with one of its four sites unexecuted) and not applicable at $L = 20$; L3 not met as written, because the open pusher at $L = 20$ holds two to three transmissible sites and the jump-based closer two; the kill criterion fired (final dominant with at most one site in 9 of 10 and 8 of 10 worlds). Every world closed, at 10,000–75,000 steps ($L = 16$) and 20,000–500,000 ($L = 20$); the heritable fraction of random cells at ten million steps is a median 0.89 and 0.95.

Controls. Stage H, well-mixed pairing (the partner drawn uniformly from the soup) at $L = 16$, ten worlds, seeds 3001–3010; pre-registered: the open phase as heritable as on the lattice (met) and closure sooner (not met: 7 of 10 worlds against 20 of 20 by 300,000 steps); the first replicator met its own kind in 57% of encounters on the lattice and 11% when mixed. Stage I, lethal tar: a zero byte fetched as an opcode halts the pair for the rest of the encounter, at $L = 16$, ten worlds, seeds 4001–4010, with the culture and partner tests under the same rule; the heritable fraction of random cells rose from 0 at 20,000 steps to 0.41 at 30,000 and 0.88 at 50,000 (medians over worlds).

BFF. cubff default semantics verified from source: two 64-byte tapes concatenated, pointer and heads at byte 0, ten instructions, every other byte a no-op, unmatched brackets halt, 2^{13} steps per encounter, 2^{17} programs, random ordered pairs each epoch, mutation 2^{-12} per byte per epoch. Switches: `ip_wrap` (pointer wraps modulo 128); `literal` (byte 0x50 = `P a b`: write `a` at `head1-1` and `b` at `head1-2`, then `head1 -= 2`, skip the operands); `nohalt` (unmatched brackets are no-ops); `halt_p` (an unmatched bracket halts with probability `p`, decided by a hash of the pair, the step and the seed); `lit_rep` (the literal instruction writes `r` copies of its word per execution). Our GPU kernel reproduces cubff’s validation replicator and matches a CPU reference on randomised tests. 24 seeds each for the published and wrapping variants, 12 for each literal variant, 12 for the published rule with no-op brackets (run after review; pre-registered: every first replicator loop-bearing, met 7 of 7; a transition in at least 9 of 12, not met at 7; pointer-closed first replicators in every world, not met in one), and 6 for each of $p = 0.01, 0.03, 0.1, 0.3$ and $r = 2, 3$ under wrap + literal; events every 64 epochs; high-order entropy with brotli quality 2 as in ref. 16. Dial predictions, registered before the run: D1-1, the open population persists (share $\geq 10\%$ at 16,384 epochs) in at least 4 of 6 soups at $p = 0.01$ (met, 6 of 6) and in at most 1 of 6 at $p = 0.3$ (not met: the share falls below 1% at 832–896 epochs in 6 of 6 and recovers to 0.10–0.11 in 5 of 6, with 0.31–0.44 of random tapes heritable); D1-2, the epoch of collapse falls monotonically with p within a factor of three of $1/p$ (not met: no collapse at $p \leq 0.1$, 896 at $p = 0.3$, 192 at $p = 1$); D1-3, no closure by control flow at any p (met); D2-1, inflow at most 1 bit at $r \geq 2$ (met: 0.00 bits in 12 of 12 soups at $r = 2$ and 3, against 8.00 in 12 of 12 at $r = 1$); D2-2, the first replicator still the loop-free, pointer-open tiling (met, 12 of 12); D2-3, collapse slower or absent at $r \geq 2$ (met: absent, with an all- P share of 22–31% at $r = 2$ and 4–7% at $r = 3$ at epoch 1,024 against 0–0.2% at $r = 1$, and every random tape heritable at 16,384 epochs).

Searches. Two-byte census: all 65,536 words tiled to 16 bytes, culture test (32 partners) and partner test (256), plus the deterministic fixed-point test against the zero partner. BFF straight-line search: 16,447,860 programs (prefix ≤ 3 instructions + tiling unit of period 2–5 with a copy instruction), 2 random partners, wrapping pointer, alphabet prefilter then exact similarity, full culture test for survivors. Closed-replicator search: all 1,309,528 periodic tilings of $\{P, [,], \text{no-op}\}$ up to period 10 under the literal, wrapping, no-halt rules, three partners each.

Pre-registration and stages. Stages D, F and G, the BFF cells and every stage added in revision (K, L, M, the mutational scan, the pair invasion, the traced execution and the dials) were pre-registered with predictions and kill criteria before launch; every amendment is dated in the change log. Misses are reported in the text: the heritable-fraction thresholds of G1(c) failed at $L = 50$; the syntactic loop criterion of G2 omitted hardware loops (6 of 20 and 3 of 20 syntactic against 19 of 20 and 8 of 20 behavioural); the expectation of 15 of 20 closures at $L = 32$ was not met (12); the mechanism first claimed for an open BFF replicator under a wrapping pointer was falsified by the harness test before any soup ran. Exploratory stages: A (630 runs, $L = 16$: seven ablations $\times$ budgets $\{32, 128, 512\} \times$ mutation $\{1/4, 1/16, 1/64\}$, ten seeds, 300,000 steps), B (640: tape lengths 4–100 under two ablations and three budgets), C (490: thirteen ablations at $L = 16$, seeds 101–110, with one-million-step nulls), E (1,230: the size axis under four control arms, budget and mutation sweeps at $L = 100$). Confirmatory stages: D (280: the $L = 9$ reversal, seeds 1001–1020), F (210: padded pair memories), G (80: closure, seeds 2001–2020, $L = 16$ and 50 for 300,000 steps, $L = 20$ and 64 for one million), K, L, M as above.

Code and data availability. The source code (simulator, assays, analysis and figure scripts), every condition file, every raw run summary, every generated table and the pre-registration documents with their dated change log are in the project repository, to be archived with a DOI on acceptance; each figure is produced from the generated tables by one script, and each number in the text is traceable to a named table.

Figure legends

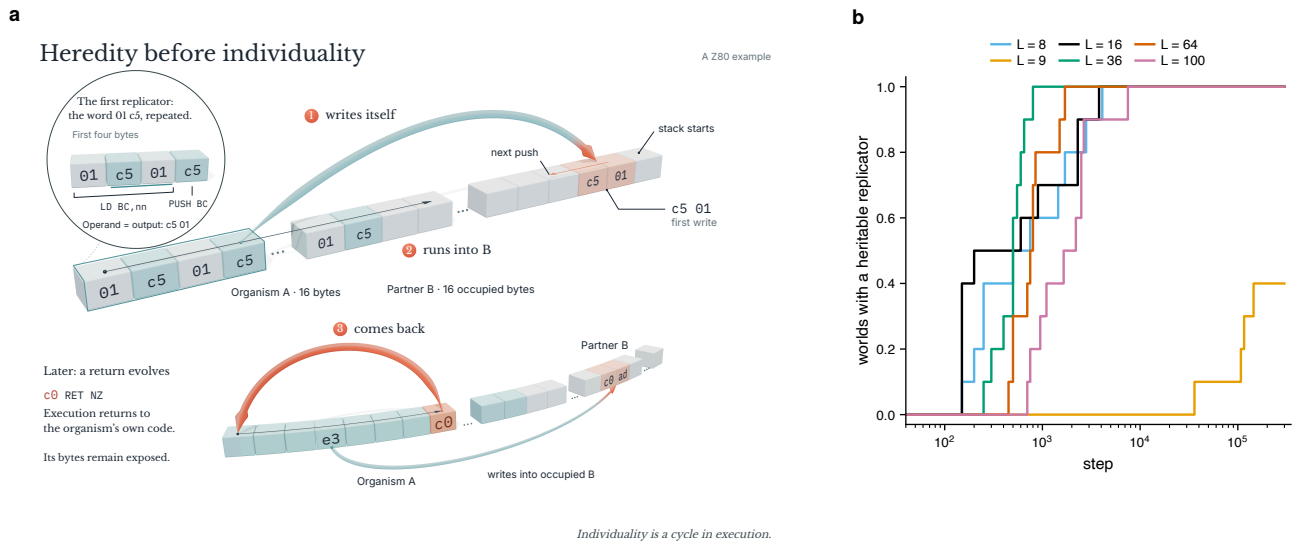


Fig. 1 | The first replicator and its closure. **a**, The first replicator and its encounter. Organism A (teal, 16 bytes) and partner B (grey, 16 bytes) share one 32-byte memory; execution starts at A's first byte and moves right, the stack pointer starts at B's last byte and moves two bytes left with every push. The first replicator is the word 01 c5 repeated: LD BC, nn loads its own next two bytes, c5 01, and PUSH BC writes exactly those two bytes at the stack pointer (operand = output), so the partner fills from its far end with the organism's own word (1) while execution runs on into the partner (2). Later a return evolves (3): execution revisits the organism's own code, and its bytes remain as exposed as before. **b**, Kaplan–Meier fraction of worlds with a heritable replicator by step for L = 8–100 (Stage E, 10 worlds per length, 300,000 steps).

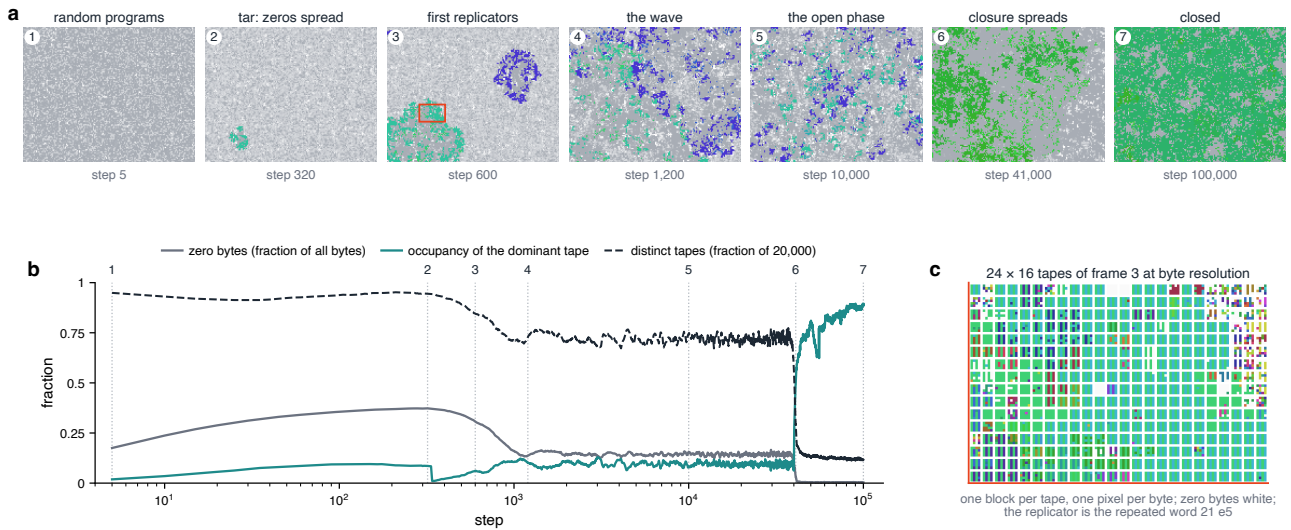


Fig. 2 | One world, watched. Seed 2002 at $L = 16$, run with dense snapshots for Supplementary Video 1. **a**, The lattice (160×125 tapes, one pixel per tape) at seven steps. The four most common tapes with at least 20 copies are coloured by the mean colour of their bytes (the colour code of **c** and of the video); every other tape is grey, lighter the more of its bytes are zero, so the tar shows as bleaching. Zeros spread first (frame 2: 37% of all bytes at step 320); the first replicators appear as patches of their own copies (3; the teal patch is the word 21 e5, the blue one 01 c5, both load-push words); they sweep the lattice (4) and compete for 40,000 steps among pockets of zeros (5); a new class carrying a conditional return (c0, RET NZ) spreads from step 41,000 (6) and takes over while the zeros disappear (7): at step 100,000 the dominant tape occupies 89% of the soup, 2,290 distinct tapes remain of 20,000, and 0.4% of bytes are zero. **b**, The same world's zero-byte fraction, dominant-tape occupancy (cells within Hamming distance 4 of the most common tape) and number of distinct tapes against step; dotted lines, the frames of **a**. **c**, The 24×16 tapes inside the red rectangle of frame 3 at byte resolution (one 4×4 block per tape, one pixel per byte, zero bytes white): the replicator's tapes are the repeated word; the rest is random code and tar.

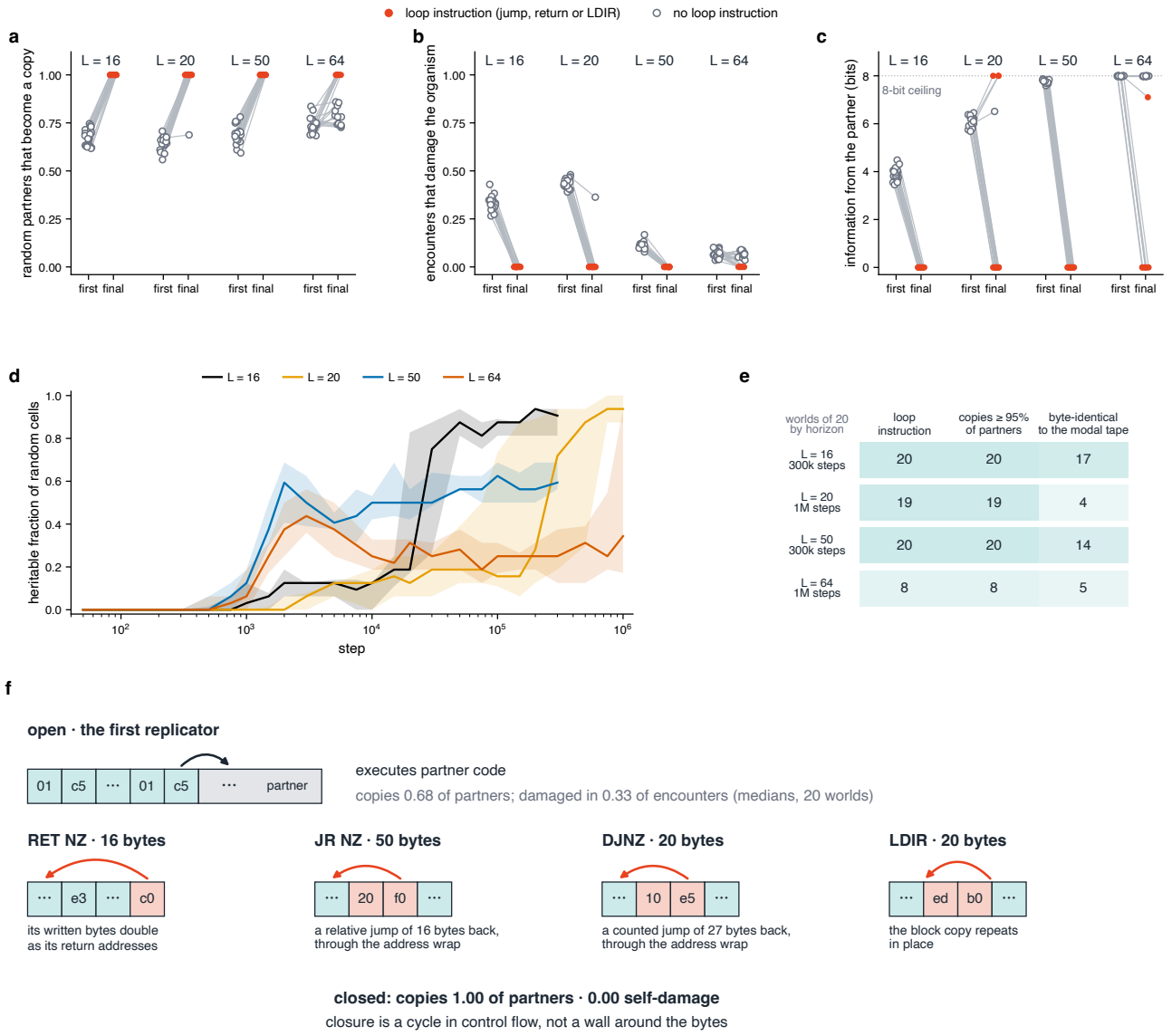


Fig. 3 | The first replicator is open, the successor closed. **a**, Fraction of 256 random partners that become a $\geq 75\%$ copy of the first heritable replicator (first) and of the final dominant (final) of every Stage G world, grouped by tape length (20 worlds per L; grey lines join the same world; open circles, no loop instruction; filled, jump, return or LDIR). **b**, Fraction of encounters in which the organism loses $\geq 25\%$ of its bytes. **c**, Information the offspring carries about the partner, $H(o | x)$ in bits over 256 random partners (dotted line, the 8-bit ceiling of 256 partners). **d**, Heritable fraction of 16 random cells against step (median and interquartile range over worlds). **e**, Worlds (of 20) whose final dominant carries a loop instruction, copies $\geq 95\%$ of partners, and equals the modal final tape byte for byte, by tape length and horizon. **f**, Control flow of the first replicator and of four closed successors, schematically (bytes abbreviated; the instruction that closes the cycle is shaded and its return drawn as a vermilion arc): the pusher's pointer runs on into the partner, whereas RET NZ returns to addresses formed from the organism's own written bytes, JR NZ and DJNZ jump back (through the 16-bit address wrap), and LDIR repeats in place. Medians of the partner test over Stage G worlds: the first replicator copies 0.68 of partners and is damaged in 0.33 of encounters; every closed successor copies 1.00 with 0.00 self-damage. Exact traces, Extended Data Fig. 3.

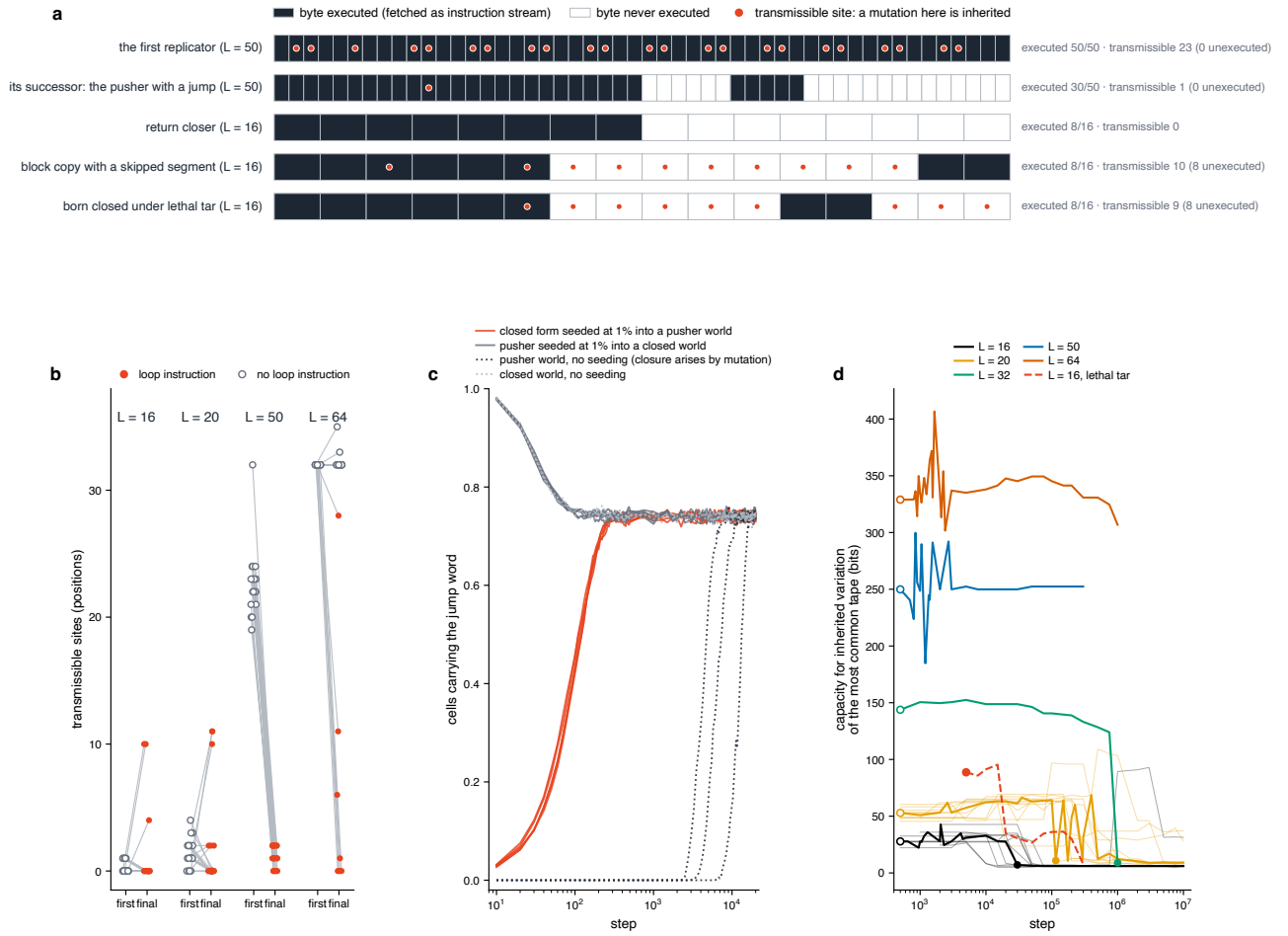


Fig. 4 | Closure wins the competition and inherits nothing. **a**, Where variation lives: for five genomes (the first replicator and the closed successor of one L = 50 world, the return design of L = 16, a block-copy tiling, and a lethal-tar final with a copied-but-unexecuted stretch), each byte is marked executed if the traced executor fetched it as instruction stream in any of 64 encounters, and transmissible if the mutational scan found at least half of its alternative values both heritable and transmitted; the pusher’s transmissible bytes are executed operands, the closers’ are bytes they copy but never run. **b**, Transmissible sites of the first replicator (first) against the final dominant (final) of every Stage G world, by tape length (grey lines join the same world; filled, loop instruction; open, none). **c**, The matched pair at L = 50: share of cells carrying the jump word 20 ± 0 against step when the closed form is seeded at 1% into a world of its ancestor (five soups), when the ancestor is seeded into a world of the closed form (five) and in ancestor-only worlds with no seeding (three), in which the jump arises by mutation. **d**, Capacity for inherited variation of the most common tape against step ($\sum \log_2(1 + 255 f)$ over positions, bits), median over worlds: Stage G at L = 50 and 64, Stage K (L = 32), Stage I (L = 16, lethal tar) and the ten-million-step worlds of Stage L (L = 16 and 20), whose individual worlds are drawn as thin lines. Open marker, the first heritable dominant; filled marker, the first snapshot at which at least half of the dominants are pointer-closed. Capacity falls when closure arrives and stays low to ten million steps; at L = 20 the median alternates between open and closed worlds while closure spreads (10^5 – 10^6 steps). The thin peaks near 100 bits are the transient genomes of three worlds, block copiers whose transmissible sites are mostly never executed, each later displaced by a design with fewer sites or none.

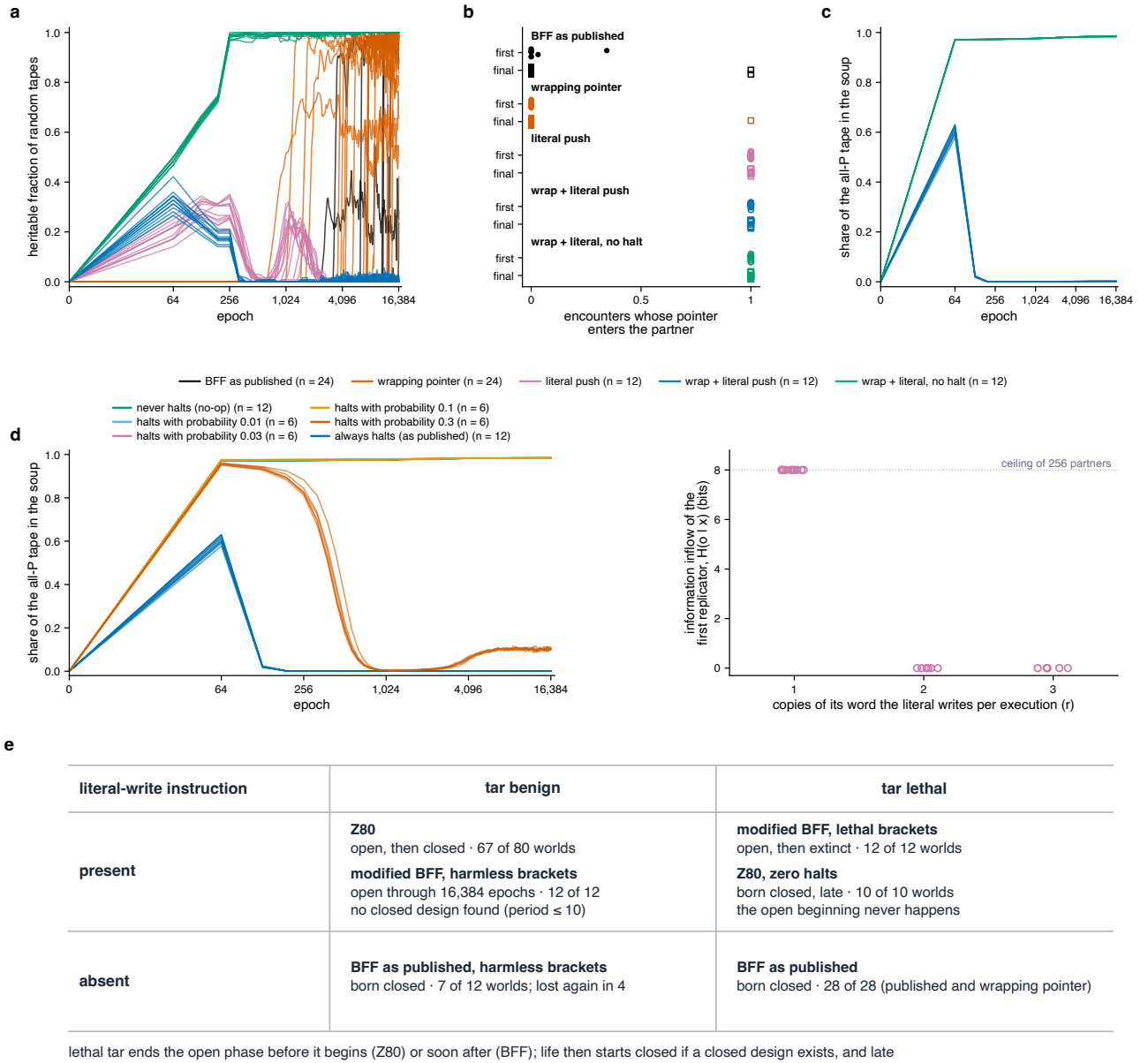
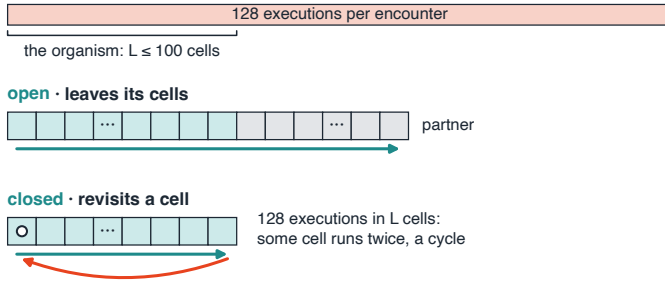


Fig. 5 | Two properties of the substrate decide the beginning. **a**, Heritable fraction of 32 random tapes against epoch for every BFF soup, by variant (as published, $n = 24$; wrapping pointer, $n = 24$; literal push, $n = 12$; wrap + literal push, $n = 12$; wrap + literal with unmatched brackets as no-ops, $n = 12$); the epoch axis is linear to 64 and logarithmic beyond, here and in **c**, **d**. **b**, Fraction of culture-test encounters in which the pointer enters the partner, for the first replicator (circles) and the final dominant (squares) of every life-producing soup, grouped by variant (open symbols, no loop instruction; filled, loop instruction). **c**, Share of the all-P class against epoch under wrap + literal (blue) and wrap + literal with no-op unmatched brackets (green), 12 worlds each (zero when the class is outside the ten largest): the same organism collapses when unmatched brackets halt the pointer and persists when they are no-ops. **d**, The two dials. Left, share of the all-P tape against epoch when an unmatched bracket halts the pointer with probability $p = 0$ (no-op, green), 0.01, 0.03, 0.1, 0.3 and 1 (as published, blue); 6 soups per intermediate p , 12 at the ends: the open replicator holds the soup at $p \leq 0.1$, collapses to a tenth of it at $p = 0.3$ and to a 0.2% minority at $p = 1$. Right, information inflow of the first replicator, $H(o|x)$ (bits) over 256 random partners, when the literal writes $r = 1, 2$ or 3 copies of its word per execution (one point per soup; open symbols, no loop): at $r \geq 2$ a copy completes within one pass and the offspring carry nothing of the partner although the pointer still enters it. **e**, Map of the substrates by two properties, a literal-write instruction (present or absent) and the lethality of the tar to an open replicator, with the observed order of events and world counts in each cell: with a literal channel and benign tar, life begins open and closes (the Z80, its 8080 subset at $L = 16$, and BFF with no-op brackets, which stays open); with a literal channel and lethal tar, the Z80 skips the open beginning and starts closed and late, and BFF's brief open beginning ends in extinction; without a literal channel BFF is born closed, and with harmless brackets loses the replicator again in 4 of 7 soups.

a



b

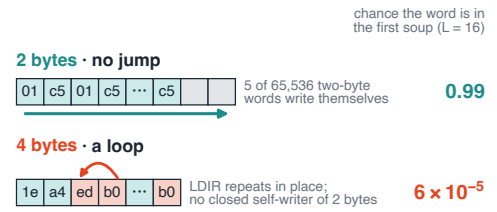
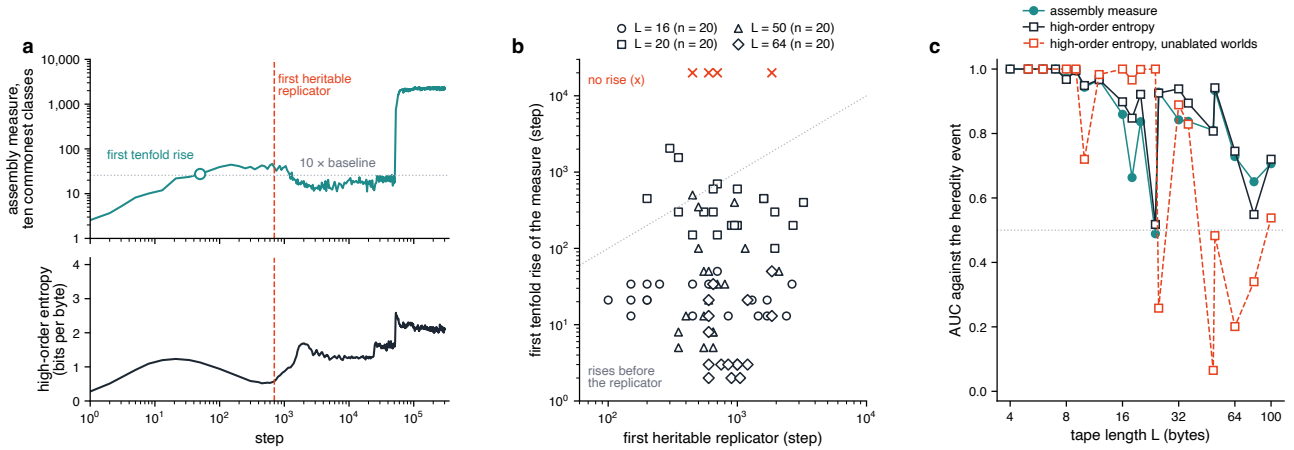
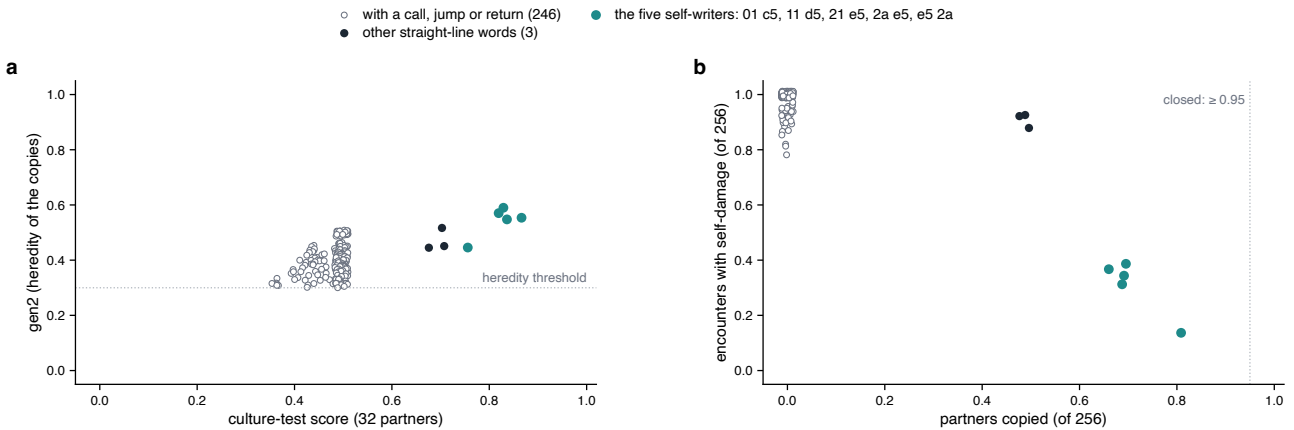


Fig. 6 | What is proved. **a**, Theorem 2 in the form that holds for the Z80 soups. An encounter executes 128 instructions, an organism has at most 100 cells, and no instruction stops execution inside it, so a program whose pointer never leaves its own cells (closed) must execute some cell twice, a cycle, whereas a program without one runs on into the partner (open). In BFF the same conclusion follows from the cost of a copy, at least 129 executions (Theorem 1). **b**, Proposition 3 with the count of Proposition 4. Of the 65,536 two-byte words, exactly five write themselves into an empty partner, all load–push pairs without a jump and therefore open, and no closed self-writer has period one or two; the smallest closer observed is the four-byte block copy 1e a4 ed b0 (LD E,n ; LDIR), whose LDIR repeats in place. Right, the chance that a given word is present among the 20,000 random tapes of length 16 at step 0: 0.99 for a two-byte word, 0.017 for a three-byte word and 6×10^{-5} for a four-byte word. Proofs, the closure-window model and the lemma on information inflow are in the Supplementary Information.

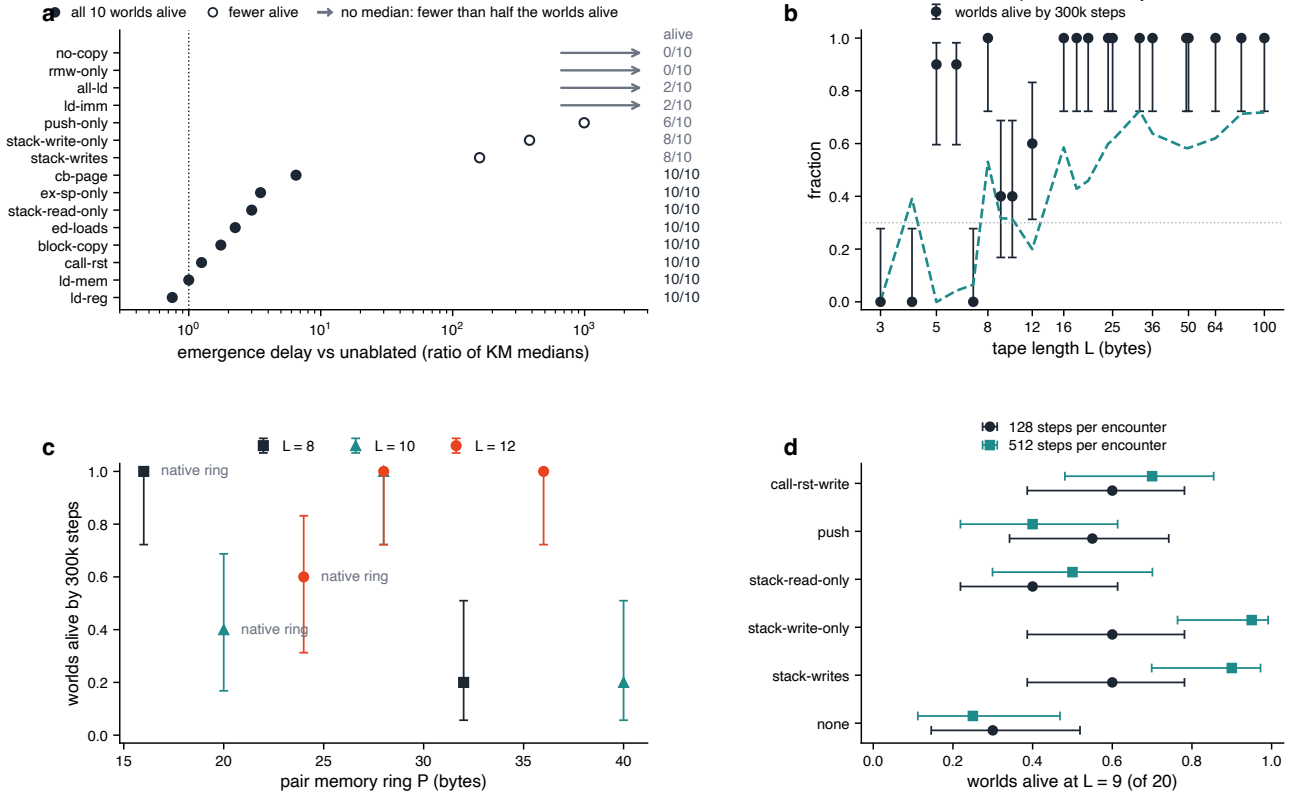
Extended Data



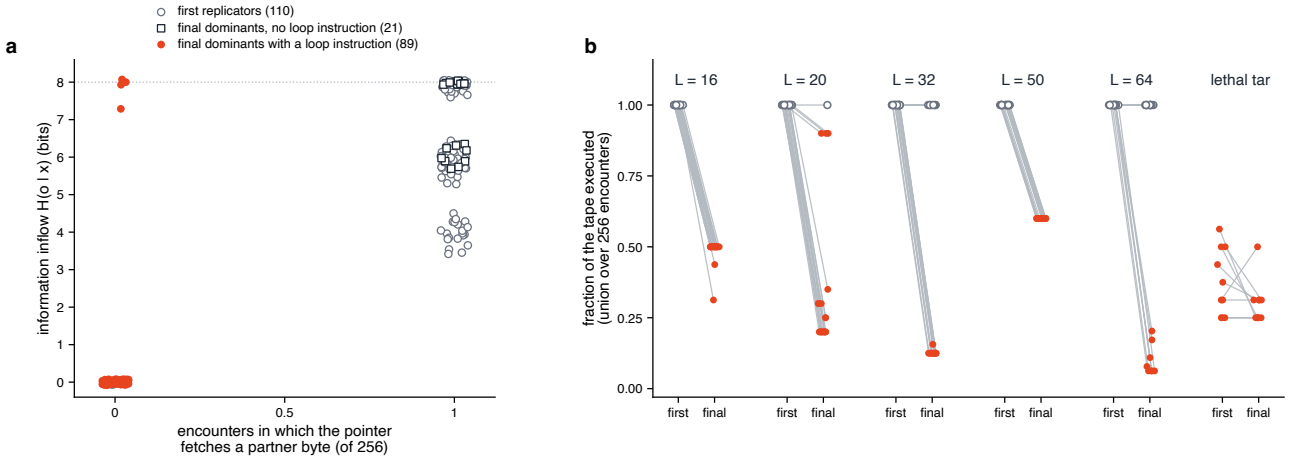
Extended Data Fig. 1 | Two detectors of life against the culture test. **a**, The assembly measure over the ten most common classes (a Re-Pair upper bound on the assembly index in Sharma et al.'s formula; top) and the high-order entropy of ref. 16 (bottom) against step for one world ($L = 16$, seed 2001): the measure rises tenfold above its random-soup baseline at step 50 (circle), 650 steps before the first heritable replicator (dashed line), while the soup is filling with tar. **b**, Step of the measure's first tenfold rise against the step of the first heritable replicator for the 80 Stage G worlds (dotted line, equality; crosses at the top edge, worlds without a rise). **c**, Area under the ROC curve against the heredity event by tape length: the assembly measure and the high-order entropy over all Stage E arms (snapshot steps), and the high-order entropy in unablated worlds alone (recorded samples, dashed), which is near chance or inverted at six of the eight lengths from 25 to 100 bytes; dotted line, chance.



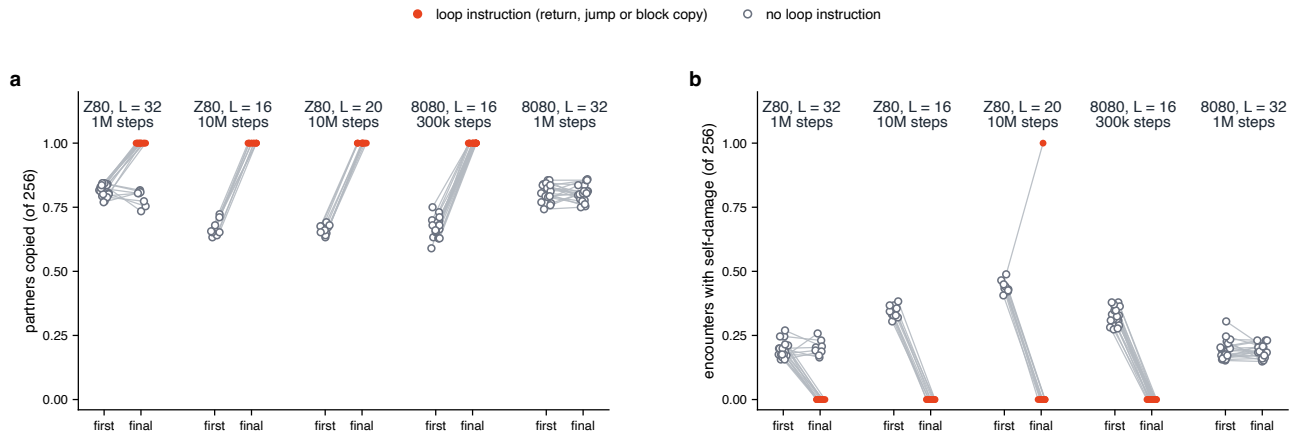
Extended Data Fig. 2 | Every two-byte word. All 65,536 two-byte words were tiled to 16 bytes and tested; shown are the 254 that pass the heredity threshold ($gen2 \geq 0.3$). **a**, Culture-test score over 32 random partners against $gen2$. The 246 words that contain a call, jump or return (open circles) pass by smearing return addresses into the partner; the five load-push words (teal) are exactly the words that write themselves into an all-zero partner (Proposition 3); the three other straight-line words (black) are their push-first rotations. **b**, The partner test over 256 random partners: fraction of partners that become $\geq 75\%$ copy against the fraction of encounters in which the word loses a quarter or more of its own bytes. No word copies into 0.95 or more of partners, and none of the 246 produces a single 75% copy.



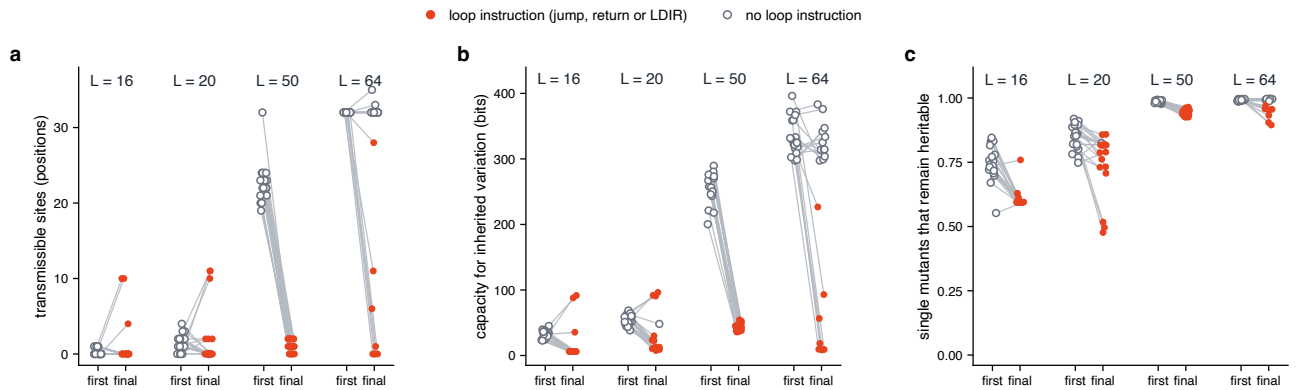
Extended Data Fig. 3 | What the instruction set must provide. **a**, Delay of emergence relative to the unablated soup (ratio of Kaplan–Meier medians, 128 steps, mutation 1/16, seeds 101–110) for each ablation; filled, all ten worlds alive; open, fewer; arrows, no ratio because fewer than half the worlds produced a replicator; right margin, worlds alive by 300,000 steps. **b**, Worlds alive by 300,000 steps against tape length (Wilson 95% intervals) and the first replicator's heritability in isolation (dashed; dotted line, the 0.3 threshold). **c**, Worlds alive against pair-memory ring length P at L = 8, 10, 12 (Stage F; the native ring is 2L): padding the ring without changing the organism removes the dead zone at L = 8–12. **d**, Worlds alive at L = 9 (of 20) by ablation at 128 and 512 steps per encounter (Stage D).



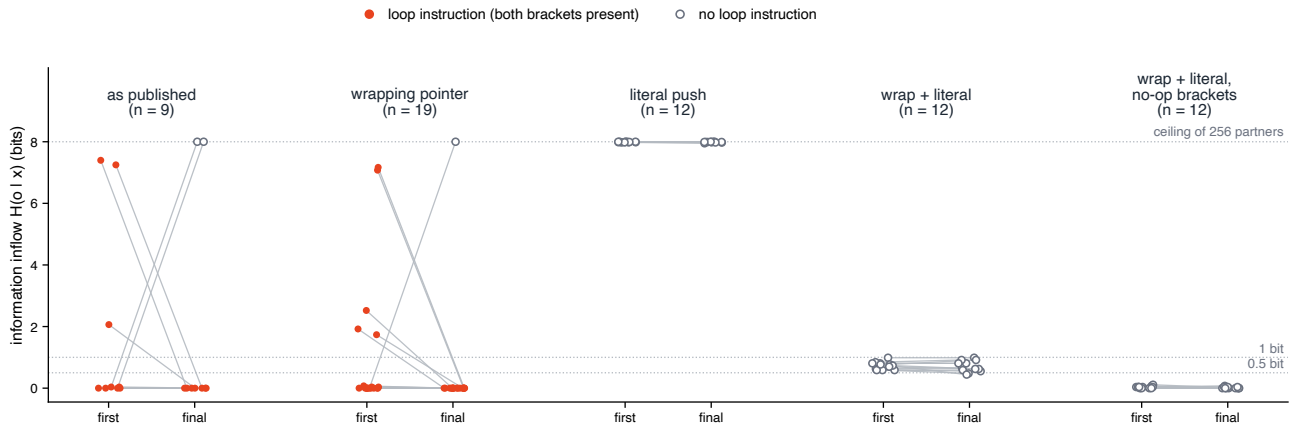
Extended Data Fig. 4 | Pointer confinement, information inflow and the executed part of the tape. Every first replicator and final dominant of Stages G, K and I (110 of each), run against 256 seeded random partners on a traced copy of the executor. **a**, Fraction of encounters in which the pointer fetches a partner byte against the information the offspring carry about the partner, $H(o|x)$ (both jittered). The 100 first replicators born under benign tar enter the partner in every encounter and carry 3.4–8 bits; the ten born under lethal tar and all 89 loop-bearing final dominants never enter it, and 85 of the 89 carry no inflow, the other four keeping one to three partner bytes (7–8 bits); the 21 final dominants without a loop instruction are open pushers. **b**, Fraction of the tape fetched as instruction stream in any of the 256 encounters, first replicator against final dominant per world (grey lines join the same world; filled, loop instruction). The pusher executes its whole tape; the closers execute part of it and regenerate the rest.



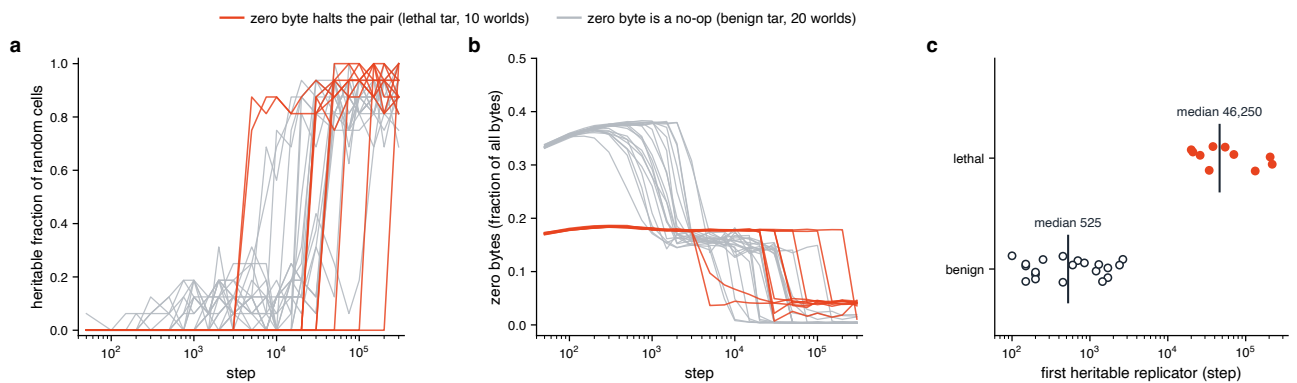
Extended Data Fig. 5 | Closure across lengths, horizons and instruction sets. First heritable replicator (first) and final dominant (final) of every world of the stages added in revision, partner test over 256 random partners (grey lines join the same world; filled, loop instruction). **a**, Fraction of partners copied. **b**, Fraction of encounters with self-damage. Z80 at the aligned length $L = 32$ (Stage K, one million steps): 12 of 20 worlds close, all by block copy. Z80 at $L = 16$ and 20 for ten million steps (Stage L): 20 of 20 close. The 8080 subset (Stage M): at $L = 16$ all 20 worlds close on the return design `ad e3 21 e3 21 c0 ad c0`; at $L = 32$, where the instruction set expresses no closer, none of 20 does. One $L = 20$ final dominant (seed 6003, a block copier at 0.3% of the soup) damages itself in every encounter while copying every partner.



Extended Data Fig. 6 | Closure trades variation for fidelity. Single-byte mutational scan of the first heritable replicator (first) and the final dominant (final) of every Stage G world, grouped by tape length (grey lines join the same world; filled, loop instruction; open, none). **a**, Transmissible sites: positions at which at least half of the tested alternative bytes leave the mutant heritable and are carried into at least half of its copies. **b**, Capacity for inherited variation, $\sum \log_2(1 + 255 \times \text{fraction of transmissible values})$ over positions, in bits. **c**, Robustness: fraction of single mutants that remain heritable ($\text{gen2} \geq 0.3$). All 255 alternative bytes per position at $L = 16$ and 20, 32 seeded values at $L = 50$ and 64; 32 partners per generation.



Extended Data Fig. 7 | Information inflow in the second machine. $H(o | x)$ over 256 random partners for the first replicator and the final dominant of every life-producing BFF soup, by variant (grey lines join the same soup; filled, both brackets present, a loop), with the 8-bit ceiling of 256 partners and the pre-registered 1-bit and 0.5-bit thresholds. Without a literal channel (as published, and with a wrapping pointer) every first replicator carries a loop and most carry no inflow; the literal push without a wrapping pointer sits at the ceiling; under a wrapping pointer the literal tiling floods the ring and carries less than 1 bit, with no-op brackets less than 0.1, although its pointer enters the partner in every encounter.



Extended Data Fig. 8 | Lethal tar in the first machine. Ten pre-registered worlds at $L = 16$ in which a zero byte fetched as an opcode halts the pair (Stage I, vermilion) against the twenty benign-tar worlds of Stage G at the same length (grey). **a**, Heritable fraction of 16 random cells against step. **b**, Zero-byte fraction against step. **c**, Step of the first heritable replicator in each world, with medians (46,250 against 525). Every lethal-tar first replicator is a block-copy replicator with a hardware loop, contains no zero byte, and copies 1.00 of partners with 0.00 self-damage from its first appearance.

Extended Data Table 1 | Final dominants of the confirmatory closure stage, by the instruction that closes the cycle. For each tape length, the 20 final dominants of Stage G grouped by the control-flow or block-repeat instruction they carry, with the most common tape of each class (first eight bytes) and the number of worlds that end on it byte for byte, and the medians of the partner test over 256 random partners.

L	instruction that closes the cycle	worlds	most common final tape in the class (worlds)	partners copied	self-damage
16	conditional return (RET NZ)	17	ad e3 21 e3 21 c0 ad c0 ... (17)	1.00	0.00
16	block copy (LDIR)	2	1e 50 f9 c3 8e 4e f6 b5 ... (1)	1.00	0.00
16	absolute jump (JP nn)	1	50 5e c3 eb 12 ee 8a 51 ... (1)	1.00	0.00
20	block copy (LDIR)	13	1e 04 ed b0 1e 04 ed b0 ... (4)	1.00	0.00
20	relative jump (DJNZ)	3	21 e5 21 e5 21 4e 10 e5 ... (2)	1.00	0.00
20	block copy (LDDR)	2	1e 9c 10 f6 6b c2 03 33 ... (1)	1.00	0.00
20	none: the open pusher	1	21 e5 21 e5 21 e5 21 e5 ... (1)	0.69	0.36
20	relative jump (JR)	1	21 e5 21 e5 21 fd 20 e5 ... (1)	1.00	0.00
50	relative jump (JR)	20	01 c5 01 c5 01 c5 01 c5 ... (14)	1.00	0.00
64	none: the open pusher	12	11 d5 11 d5 11 d5 11 d5 ... (5)	0.78	0.06
64	block copy (LDIR)	5	84 5e ed b0 84 5e ed b0 ... (3)	1.00	0.00
64	block copy (LDDR)	2	b8 4b 2c cf ef ec ae 83 ... (1)	1.00	0.00
64	absolute jump (JP (HL))	1	d5 d5 11 73 e9 d5 11 d5 ... (1)	1.00	0.00

Extended Data Table 2 | Instruction-set ablations. Each ablation replaces the listed instruction families or mnemonics by no-ops through a prefix-aware fetch hook; the count is of opcodes on the base, CB and ED pages (indexed DD/ED forms follow their base opcode); a leading minus removes a family from the set. i8080 is the 8080 subset of Stage M.

ablation	opcodes made no-ops	patterns (instruction families or mnemonics)
none	0	(none)
block-copy	4	block-copy
stack-writes	46	stack, ex, call-ret, rst
stack-write-only	22	PUSH, EX (SP),HL, CALL, RST N
stack-read-only	24	POP, RET, EX DE,HL, EXX, EX AF,AF'
push-only	8	stack
push	4	PUSH
ex-sp-only	1	EX (SP),HL
call-rst	34	call-ret, rst
call-rst-write	17	CALL, RST N
ld-mem	31	ld8-mem, ld16-mem
ld-imm	11	ld8-imm, ld16-imm
ld-reg	54	ld8, ld-special
all-ld	96	ld8, ld8-imm, ld8-mem, ld16-imm, ld16-mem, ld-special
ed-loads	8	8 ED-page opcodes
cb-page	256	rotate, rotate-mem, bit, bit-set, bit-set-mem, -RLD, -RRD
no-copy	112	ld8, ld8-imm, ld8-mem, ld16-imm, ld16-mem, ld-special, stack, ex, block-copy
rmw-only	46	writes-mem, -incdec-mem, -rotate-mem, -bit-set-mem
i8080	342	bit, bit-set, bit-set-mem, rotate, rotate-mem, EX AF,AF', DJNZ, JR, EXX, 78 ED-page opcodes

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Supplementary Information: theorems and proofs

What kind of statement each item is: a **Theorem** has a pen-and-paper proof and uses no simulation; a **Proposition** is an exact statement decided by finite computation, with the computation described; a **Model** is a result about a stylised model with stated assumptions, not about the simulator. Empirical premises are marked as such, and every number quoted is from the generated tables behind the main text.

Notation. An encounter executes a program held in a shared memory. The organism A occupies addresses 0 to $L - 1$ and the partner B the next L addresses; in the Z80 soups the $2L$ addresses form a ring, in BFF the 128 addresses do not wrap. Execution starts at address 0 with a fixed initial machine state (registers zero; the Z80 stack pointer at the last byte of B) and lasts at most T instruction executions ($T = 128$ in the Z80 soups, 2^{13} in BFF). A *control transfer* is a jump, call, return, repeat-in-place or halt; a *straight-line program* executes none, so that under default flow its pointer advances past each executed instruction. The *no-op context* is the partner all of whose bytes are no-ops (zero in the Z80, where byte 0

is NOP; zero in BFF, where every byte outside the ten instruction characters is a no-op). A tape x is a *self-writer* if its encounter with the no-op context ends with B holding a copy of x (up to a cyclic shift, as in the culture test), and a *full self-replicator in context e* if the encounter (x, e) ends with B holding a copy of x . It is *closed* if its pointer executes no address of B in any context and *open* if it does in some context. A value is *prescribed* if it is determined by x alone, the same for every partner.

Theorem 1 (no one-pass replicator in BFF). *Setting.* BFF as published: a 128-byte pair; the instruction pointer p and the two data heads $h0$ and $h1$ start at address 0; p advances by one after every instruction unless a bracket jumps; $< >$ and $\{ \}$ move $h0$ or $h1$ by one cell; $+$ — change the byte under $h0$ by one; $.$ copies the byte under $h0$ into the cell under $h1$ and $,$ the byte under $h1$ into the cell under $h0$; every other byte is a no-op; execution ends when p leaves the pair, on an unmatched bracket, or after 2^{13} steps. *Statement.* Let A be a 64-byte tape with at least two distinct byte values and B a partner that initially holds no byte of the copy in place. Any execution that leaves B holding a copy of A runs at least 129 instructions. Hence no straight-line execution, which runs at most 128 (one per address), produces a copy into such a partner, and a program that copies into every partner, as the culture test requires, is not straight-line; a constant tape produces no copy either. Writes, increments and moves are counted as disjoint sets of instructions below: w counts $.$ and $,$, p counts $+$ and $-$, v counts the four moves. *Proof.* Let the copy have $D \geq 2$ distinct values. Each of the 64 cells of B must receive a write of a prescribed value. $+$ and $-$ applied to a cell holding an unknown partner byte give an unknown byte, so a prescribed value reaches a cell only through $.$ or $,$, or through $+$ or $-$ applied to a cell that already holds a prescribed value; therefore at least $w \geq 64$ write instructions run. A write lands on the cell under the writing head, so the 64 target cells are among the positions the heads occupy; both heads start at address 0 and each move changes one head's position by one, so after v moves the heads have occupied at most $1 + v$ distinct positions. The first write of each of the D values must read it from a cell that holds it; before any cell of B holds a prescribed value only cells of A do, and a cell of A acquires a value it did not hold initially only by a write from a cell that held it or by an increment; so, if p increments run, the heads also occupy at least $D - p$ distinct cells of A that held distinct values initially. Target cells lie in B and these source cells in A , so $64 + D - p \leq 1 + v$. The instruction count is therefore at least $w + v + p \geq 64 + 63 + D \geq 129$. If $D = 1$ the tape is one repeated byte and, as a program, one repeated letter: a move letter never writes, a write letter with both heads at address 0 copies a cell onto itself, and $+$ or $-$ never write into B . ■ *Corollary 1.1.* Every self-replicator of BFF runs at least 129 instructions within 128 addresses, so it executes some address twice: it carries a loop. Whether the loop also keeps the pointer out of B is a separate, measured fact (0 of 28 first replicators were open).

Theorem 2 (closure requires a cycle). *Hypothesis (H1), default flow.* In the absence of an executed control transfer the pointer advances past each instruction, so that a straight-line program runs from the organism's region into the partner's. *Statement.* Under (H1), a closed full self-replicator of L bytes executes some address at least twice, provided either (i) a copy of it costs more than L instruction executions, or (ii) execution cannot end inside the organism's region before a budget $T > L$ is spent. *Proof.* A closed execution visits only the L addresses of A . If no address repeats, it comprises at most L instruction executions. Under (i) these cannot complete the copy; under (ii) the budget forces a further execution, which can only be at an address already executed. ■ *The machines.* BFF satisfies (i) by Theorem 1: a copy costs at least 129 executions, more than the whole pair has addresses. The Z80 soups satisfy (ii): the budget is 128 executions for tapes of at most 100 bytes, there is no exit instruction, and HALT holds the processor at its own address, which we count as a one-state cycle. *The bandwidth reading.* Condition (i) holds in any machine whose straight-line code writes fewer bytes than the addresses it consumes, since a pass over L addresses then writes fewer than L bytes. This is the form used in the main text, and it describes the replicators observed: the first replicator writes two bytes per four bytes of code (the executor trace shows eight bytes written when the pointer leaves its sixteen cells), BFF's copy instructions move one byte per instruction and need a head move per cell, and the literal push of modified BFF writes two bytes per three. It is not a machine-wide property of the Z80, in which a chain of pushes writes two bytes per byte of code; for the Z80 the theorem rests on (ii). *Corollary 2.1.* Under (H1) every straight-line program is open: with $T > L$ its pointer reaches the partner's region. A closed self-writer therefore contains an executed control transfer besides the instructions that write its content, and where a straight-line self-writer exists the closed one is longer by at least that instruction. For the Z80 the comparison is exact: by Proposition 3 the self-writers of period one or two are exactly five straight-line load–push words, so the smallest closed self-writer has period at least three (the closed replicators observed have period four or more).

Proposition 3 (the two-byte self-writers of the Z80; computer-assisted). Among the 65,536 two-byte words tiled to 16 bytes and executed for 128 instructions as A against the all-zero partner, exactly six leave B holding a tiling of the word: the trivial 00 00, which B already holds, and the five load–push words 01 c5, 11 d5, 21 e5 (LD rr, nn ; PUSH rr) and 2a e5, e5 2a (LD HL, (nn) ; PUSH HL), decided by exhaustive execution of all 65,536 words. None contains a control transfer, so each runs on into the partner: every two-byte self-writer of this machine is open, and no tiling of period one or two is closed. The stochastic census agrees: the same five words copy into 0.66–0.81 of 256 random partners, none into 0.95 or more, and the 246 words that pass the heredity threshold by smearing return addresses produce no 75% copy.

Proposition 4 (open first: a count on the initial soup). In a soup of N tapes of L uniformly random bytes, the expected number of occurrences of a given k -byte word, at any of the $L - k + 1$ positions of a tape, is $N(L - k + 1) / 256^k$: for $N = 20,000$ and $L = 16$, 4.6 for a given two-byte word, 0.017 for a given three-byte word and 6.1×10^{-5} for a given four-byte word. A given two-byte word is present at step 0 with probability $1 - (1 - 256^{-2})^{N(L-1)} \approx 0.99$, a given three-byte word with probability ≈ 0.017 . *Consequence (conditional).* If the two-byte open self-writer has a positive growth rate in the soup (an empirical premise: it copies into 0.66–0.73 of random partners and invades from 1% of cells, Extended Data Fig. 7) and every closed self-writer has period at least three (Proposition 3), then an open self-writer is present from the start with probability ≈ 0.99 while a given closed design is absent with probability ≥ 0.98 (≥ 0.9999 at period four, the smallest observed), and the first replicator is open unless the open form fails to grow. This is the status of “open first”: a theorem-grade count plus one empirical premise.

Model 5 (the closure window). *Assumptions.* An open replicator population of $n(t)$ tapes; each tape-step produces a closed functional variant with probability q (a mutation placing the right bytes at the right phase: q is of order sites $\times \mu \times 256^{-m}$ for an m -byte closing motif); a closed variant, once present, grows faster than the open form and fixes with probability of order one. *Statement.* The expected number of closed variants produced while the open population lasts is $q \int n dt$, and closure occurs with probability $\approx 1 - \exp(-q \int n dt)$; the integral is the window. If the open form’s own copies are viable in the soup it creates, n stays near the carrying capacity, the window grows without bound and closure is a matter of waiting; if the soup becomes lethal to the open form before $q \int n dt$ reaches order one, the open form dies without closing. *What the experiments found.* Z80, benign zeros: closure in 20 of 20 worlds by 300,000 steps at $L = 16$ and 50, in 19 of 20 by one million at $L = 20$ and in 8 of 20 at $L = 64$, where the open form is least damaged and the motif must land in the longest tape (smallest q). BFF with the literal push and lethal brackets: n fell from $0.6 N$ at epoch 64 to $3 \times 10^{-4} N$ by epoch 256, a window of order 10^{19} tape-epochs, and 0 of 12 closures. BFF with benign brackets: the window is unbounded, yet no closure by control flow in 12 of 12, consistent with $q = 0$, since no closed heritable tiling exists in that language up to period 10 (Methods); the open form persists for 16,384 epochs. Z80 with lethal zeros (Stage I): the open form never establishes, so the window is empty from the start, and the first replicator is closed and late (median 46,250 steps against 525; 10 of 10 worlds). The model does not predict which regime a machine is in; that depends on how the open replicator’s own by-products act on it, which is what the experiments measure.

Theorem 6 (architecture decides the first replicator). *Hypotheses.* (H1) Default flow, as in Theorem 2, with execution ending when the pointer leaves the pair (as in BFF) or continuing around a ring (as in the Z80 soups). (H2) *Copy cost:* writing a copy of L bytes with at least two distinct values costs more instruction executions than the pair has addresses, $2L$ (true of BFF by the proof of Theorem 1, and of any machine whose write instructions place one byte at a head that moves one cell per instruction). (H3) *Self-consistent literal write:* an instruction that writes its own operand bytes verbatim at a write pointer that starts in, or advances into, the partner’s region, and whose operand can be its own encoding in the phase executed (`LD BC, nn ; PUSH BC` with $nn = c5\ 01$; `P a b` with $a = b = P$). *Statement.* (i) Under (H1) and either proviso of Theorem 2, every closed self-replicator contains a cycle. (ii) Under (H1) and (H3), the tiling of the literal instruction is a straight-line program, hence open, that writes its own content at the advancing write pointer; it is a self-writer wherever the encounter lets the written bytes cover the partner’s region, which holds in the Z80 (eight pushes over two passes of the $2L$ -byte ring; Proposition 3) and in BFF with P under both pointer rules (12 of 12 soups in each; the straight-line search, Methods). (iii) Under (H1) and (H2) without (H3), in a machine whose execution ends when the pointer leaves the pair, no straight-line program is a self-writer and every self-replicator contains a cycle: born loop-bearing (whether its pointer also stays out of the partner is a separate, measured property). (iv) Under uniform random initialisation over an alphabet of size σ , the expected number of occurrences of a fixed k -symbol pattern among N regions of length L is $N(L - k + 1) \sigma^{-k}$, so a minimal open pattern shorter than every closed one is present with probability higher by a factor exponential in the length difference. *Proof.* (i) is Theorem 2. (ii) The tiling executes no control transfer, so by (H1) its pointer runs into the partner’s region: it is open; its writes are its operands, by (H3) the tiling itself; completion of the copy is the per-machine fact cited. (iii) In the no-op context a straight-line program runs at most $2L$ instructions before the pointer leaves the pair, and by (H2) these cannot complete a copy; so a self-writer executes a control transfer, and since a copy costs more than $2L$ executions within $2L$ addresses, some address repeats. (iv) is linearity of expectation. ■ *What it decides.* Whether a substrate’s first replicator can be open is decided by (H3). The Z80 has a self-consistent literal write (open first, 80 of 80 worlds); BFF has none, since its instructions increment in place or copy from memory (28 of 28 first replicators loop-bearing, and 0 of 28 enter the partner by measurement); BFF with one literal-write instruction added is open first (12 of 12). The theorem says nothing about how long the open phase lasts (Model 5). *Scope.* (H1) is a property of the von Neumann architecture, shared code and data under sequential control, and holds in Core War, Tierra, Avida, the Z80 and BFF alike. (H2) is a property of the copy instructions and must be checked per machine: it holds in BFF by Theorem 1 and fails in the ring Z80, whose two-byte self-writer completes its copy on a second pass of the ring, so for the Z80 the open-first conclusion rests on Propositions 3 and 4 and on observation. For a chemistry the hypotheses are a reading, not a theorem: (H1) would

be a copier that runs off its template into adjacent material by default, (H3) a template whose content specifies its own complement without an interpreter.

Lemma 7 (closure as zero information inflow). Let the offspring of x in context e be $o = F(x, e)$, with e drawn from a context distribution, and define the inflow $\lambda(x) = I(o; e | x)$, the information the offspring carries about the context given the parent. (i) If $\lambda(x) > 0$ then $P(o = x | x) < 1$: a replicator with inflow cannot be perfectly faithful across contexts. (ii) If $P(o = x | x) = 1$ then $\lambda(x) = 0$: perfect fidelity across contexts implies zero inflow. (iii) Fano's inequality bounds the fidelity: $P(o \neq x | x) \geq (H(o | x) - 1) / \log|O|$, where O is the set of possible offspring. *Proof.* If $o = x$ with probability one given x , then o is a function of x alone and $I(o; e | x) = 0$, which is (ii); (i) is its contrapositive; (iii) is Fano's inequality with $H(o | x) \geq I(o; e | x)$. ■ *Measurement.* For a deterministic executor $H(o | x, e) = 0$, so $\lambda(x) = H(o | x)$, the entropy of the offspring over partners; the partner test estimates it as the plug-in entropy over 256 uniformly random partners (ceiling 8 bits). *Reading, with the measured qualification.* The lemma is Krakauer et al.'s individuality, the organism's future determined by its own past rather than the environment's, specialised to one generation and made measurable by intervention. Pointer closure (execution never enters the partner) is neither necessary nor sufficient for $\lambda = 0$: it is not sufficient because offspring positions the organism never writes keep the partner's bytes (7 of 28 pointer-closed BFF first replicators have $H(o | x) > 0$, three of them ≈ 7 bits from a single unwritten byte), and it is not necessary because a copier that overwrites everything it touches has $\lambda \approx 0$ although its pointer enters the partner (the wrapping literal pusher of BFF, 0.04 bits). Inflow comes from partial copying; openness of control flow is the mechanism that makes the Z80 first replicator's copies partial. Measured: the first replicator has $\lambda > 0$ in 80 of 80 worlds (median 7.0 bits) and the loop-bearing successor $\lambda = 0$ in 63 of 67 (one offspring string in all 256 contexts), the four exceptions copying all but one to three bytes (Fig. 3c).